

Week 8: An Introduction to LLMs

Key Concepts:

- **Large Language Models (LLMs):** Computer programs trained on massive amounts of text to understand and generate human-like language.
- **Natural Language Processing (NLP):** The enablement of machines to effectively interpret, generate, and interact with human language data.
- **Tokenisation:** The process by which LLMs break down text into small pieces called 'tokens' (words or parts of words) to process information.
- **Probabilistic Prediction:** LLMs work by predicting the most likely next token based on what they have already written, rather than truly "thinking".
- **Hallucinations:** A major challenge where LLMs generate facts that sound convincing but are actually incorrect or nonsensical.
- **Determinism vs. Creativity:** The balance between getting the same predictable answer every time and allowing the model to be more creative in its responses.
 - Key to striking the balance between determinism and creativity is the choice of **hyperparameters** (e.g. Temperature, Top_p)
- **Model Drift:** When a model's performance or behaviour changes over time as it is updated, leading to inconsistent results.

Techniques introduced:

- **Prompt Engineering:** The technique of crafting detailed instructions (using personas, context and specific instructions) to guide the LLM toward better outputs. Including:
 - **Grounding:** Adding external information/sources to a prompt to improve factual accuracy and minimise fabrication.
 - **Few-Shot Prompting:** Providing the model with a few examples of a task so it can learn the specific pattern you want it to follow.
 - **Chain of Thought:** Asking the LLM to "explain its reasoning" step-by-step, which helps it handle complex tasks more accurately.
 - **Prompt chaining:** Breaking complex tasks into smaller, sequential, and manageable steps, where the output of one prompt becomes the input for the next
- **Retrieval-Augmented Generation (RAG):** Connecting an LLM to specific external documents so it can provide answers based on that private data.
- **AI-as-Judge:** Using a second AI model to review and grade the quality of the response generated by another model.
- **Pinned Models:** Using a specific, fixed version of an AI model to ensure results remain consistent over time and avoid the issue of 'model drift' (see above).

Applications introduced:

- **Google AI Studio:** A browser-based environment where users can test different models and fine-tune settings like “Temperature” and “Top_p”).
- **NotebookLM:** A tool by Google that allows users to upload documents and use AI to chat with them or generate audio summaries.